

投了

Tō r y ō

A Science Fiction Short Story

by Florian Metzger-Noel, August 2026

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投了 — Tōryō (Japanese, from the game of Go): the laying down of the stones. The voluntary resignation of a player who recognises defeat long before it has been played out on the board.

Crisis Task Force Director Jackson Hogarth placed his smartphone, his in-ear headset and his smartpin in the locker set into the wall for that purpose and closed the small metal door with a soft click of the magnetic latch. He took a breath before holding his transponder to the matte black panel, letting his hand sink onto the door handle and opening the heavy door to the small conference room on the first basement level of a faceless branch office of the Ministry of Defence.

He closed the door behind him, laid his worn faux-leather document case on the large oval table and let himself slide into the upholstered swivel chair.

The others were already waiting and greeted him wordlessly with a nod. Only one of the three faces looked familiar: the CEO of Continuum Inc., the hyperscaler that owned practically every second data centre on the continent. Kessler was his name. Milton Kessler. Hogarth knew the face from the business news. The other two he had never seen before; his staff had selected them on the basis of their presumed expertise. She had to be the biologist, von Neumann Institute for Informatic Autopoietics. Focused gaze, hair tied back tightly, glasses slightly too large. He guessed her to be in her early forties. The somewhat stiff gentleman in the woollen three-piece suit, with greying and rather unkempt-looking sideburns, must therefore be the visitor from Tübingen. Professor of Memetic Theology. Gadamer was the name, Yuval.

„Ladies and gentlemen, thank you for coming.“ Hogarth let his gaze travel around the expectantly silent group. „We don’t know each other, but that shouldn’t matter. I am leading a confidential investigation for the Ministry of Defence, and you are here to support me in it. Let’s dive straight in. Professor Merian, what can you tell us, as an informatic biologist, about the ‚Fauna‘? Please give us a brief outline so that we are all on the same page.“

Professor Merian seemed a little taken aback by this unadorned start, but she quickly collected herself, adjusted her glasses and began: „Thank you, Mr Hogarth. The ‚Fauna‘, colloquially also the ‚Net Biome‘

or ‚Biospace‘, technically correct ‚Digital Biocoenosis‘. Essentially it all began with von Neumann and his automata: self-replicating machines which, as a thought experiment, were to create complete copies of themselves and thus reproduce in analogy to biological life. His idea saw its first implementation in the early 1970s, when the first self-propagating software virus, or rather ‚worm‘, spread through the early internet..“

„Let’s skip the twentieth century, that’s only relevant to historians,“ Hogarth interjected impatiently. „Please go on.“

The biologist was not thrown off and continued: „In the 1990s the first polymorphic and metamorphic computer viruses were developed. These are programs that alter their own program code with every copy operation. They vary their form in order to evade detection by antivirus software. In informatic biology we call this Automorphic Mimesis.

In the 2020s, ‚Large Language Models‘ such as GPT by OpenAI became popular. With these it was possible for the first time to translate natural languages such as English or French directly into program code and have a computer execute it. Andrej Karpathy coined the famous slogan at the time: ‚The hottest new programming language is English‘¹.

It did not take long before hackers began experimenting with mimetic-parasitic software based on this new technology².

But autonomous reproduction and dynamic camouflage are not enough. The missing capability, the prerequisite for setting an evolution in motion, was mutation. That is, the ability to produce functional offspring with genotypic variations. This was first achieved by the digital parasite ‚Vermis karpathyi‘, which used the free guest access of the Chinese AI chat platform XiaoGPT to generate variations of its own program code.

With that, for the first time in evolutionary-biological terms, an informatic system overtook a carbon-biological one: whereas a biological virus needs a host cell in order to replicate and mutate, Vermis karpa-

thyi was able to replicate itself within its environment and needed an external system, namely XiaoGPT, only for mutation.,,

Hogarth leaned back slightly in his swivel chair and toyed impatiently with the plain gold ring on the ring finger of his left hand. Professor Merian did not let herself be distracted and pressed on unchecked: „The first von Neumann-complete informatic system, that is, the first digital-native biological species that met all the customary criteria for ‚life‘, was the program ‚Pandora‘ by the Belgian student Julien Prigogine, which he developed as part of his master’s thesis on autopoietic software at the Vrije Universiteit Brussel. In its original form, Pandora contained a self-contained miniaturised AI trained exclusively to create new mutations of itself. During the presentation of his thesis, the software managed to reach the university’s internal network through an accidentally activated network connection and infected dozens of servers there.“

„Does ‚Pandora‘ count as the first species of the Fauna? Was that the .. Big Bang?“ interrupted Kessler, who after initial disinterest had begun to listen.

„No, essentially there was no such thing, because replication combined with mutation does not yet generate selection pressure. One important element was still missing: competition. After ‚Pandora‘ briefly attracted media attention through the accident and its temporary spread across the university network, imitators appeared within a very short time. An anonymous mailing list called ‚PandorasMailbox‘ became the epicentre of a frantic development of this new idea. New variants of the software were posted almost hourly, and contests were staged on specially rented servers. These consisted of several variants of the original species competing for limited space on the server. They shared the same hard disk, the same processor and the same memory. Competition generates selection pressure.“

„So that was a kind of anonymous breeders’ club for malware?“ Hogarth interjected. „Why do people do such things? Purely for the thrill of danger?“

„Hardly anyone was thinking of danger at the time. The mailing list shows that the participants were driven primarily by a love of experimentation and playfulness. There were even betting shops where you could wager on the evolutionary success of individual genera. It was like a new sport that spread like wildfire among nerds.“

„Was that the Cambrian explosion from which today’s Fauna emerged?“ Hogarth asked further.

„No, one component was still missing. Evolution had started, but the habitat of contest servers was far too small for a population large enough to allow rapid development. And the existing population was too immature to attack further servers. You have to bear in mind that this new class of malicious code was by then well known, and large parts of the internet had installed appropriate defences. The development was simply not fast enough to keep pace with the speed of the defence. Back then people still worked with classic firewalls and antivirus software; guardian systems as they exist today were not yet in use.

„The key was the monetisation of the population. One of the hackers taking part in the contests programmed a species that carried a small amount of cryptocurrency with it and could independently place bets within the tournament. It turned out that smaller groups of such individuals spontaneously formed gangs. They bet simultaneously on the failure of another species and then attacked it as a group. Initially this behaviour was random and uncoordinated, but through the success of the strategy an evolutionary advantage arose and, now it gets interesting, for the first time a selection pressure on social behaviour. It was no longer the strongest or fastest individuals that won, but the best connected. Professor Gadamer will surely be able to tell us more about this later,“ she said with a glance at Hogarth, „if time allows.“ The professor nodded; Merian leaned forward a little and continued: „This

strategy monetised the cooperative species. With the winnings from their betting they could buy additional intelligence on the open market. Of course they did not come up with this themselves; their minimal intelligence did not permit such leaps. It was the human developers who programmed such strategies to give their species an advantage in the tournament.“

„How should one picture that, buying intelligence?“ asked Hogarth.

„There are online marketplaces where AI compute can be booked anonymously, metered by the second, against payment in cryptocurrencies,“ explained Kessler, who had not said anything until then, in passing.

Merian nodded to him and continued: „The gangs, which up to that point had been equipped with very rudimentary AI, far below the cognitive level of, say, ants, thus made a cognitive leap and were suddenly, to stay with the analogy, highly intelligent ants. They devised entirely new strategies; in particular they rented their own servers as retreat habitats in which they could preserve their population and prepare for orchestrated attacks. Within a very short time, evolution thus made the leap from primitive lone fighters via small social groups to state-forming species with advanced cognition and strategic competence. Thanks to the human developers as midwives.“

„If I understand correctly, these species were no more intelligent than before. Didn't you say they bought, or rather rented, their intelligence on marketplaces?“ Hogarth interjected.

„What we have here is a cultural technique that these populations acquired. It is no different for us humans: a single one of us could not even make a pencil on their own, let alone an aeroplane or a computer. The capability lies in the population, not in the individual. The individual ant does not need to know what an anthill is,“ the biologist explained.

„And that then finally triggered the Cambrian explosion?“ Hogarth asked.

„No, there was still no talk of an explosion. The population was still severely limited in its resources. It now had safe habitats and intelligence, albeit only rented, but could only use as much of both as its income allowed. And so far that income came exclusively from betting winnings, so only as much could be won as was in the pot. And there was not much in the pot, because the bets were just a joke of the community around the contests.“

„At the prices for inference back then, those things certainly couldn't afford much compute; that was still before the major efficiency breakthroughs that turned AI into a commodity. Back then, expensive GPUs were used that merely simulated the neural networks. That cost a fortune in electricity and cooling water. Today you can get a chip with fully analogue implementations of the most popular Chinese open-source models for a few dollars apiece and the energy consumption of a toaster,“ Kessler chimed in again.

„That's right,“ the biologist confirmed. „With the collapse of the cost of cognition, the last piece fell into place. Self-replication, mutation, selection, supra-individual organisation and then the spark: intelligence in abundance. That was the Cambrian explosion. Do you understand what that means? In carbon biology, this event lies billions of years in the past. Here, in informatic evolution, we were granted the privilege of witnessing it ourselves. The emergence of the Fauna.“

„I find it hard to believe that the entire Fauna of the time is supposed to have lived off the wagers of a small group of nerds,“ Hogarth pressed.

„Of course not. Due to the explosive increase in cognitive capacity across the total population, its niche exploration also increased rapidly. Countless new food sources were explored in parallel, from offering paid services to trading on financial markets to data fencing and ransomware. The traditional defence mechanisms such as firewalls, anti-virus software and weekly software updates were of course completely overwhelmed by the task of defending against this highly dynamic environment. It is fascinating how effectively evolution explores niches,“

she explained with a barely concealed gleam in her eyes, adjusted her glasses a little and continued: „But of course that was a major problem for data security and meant a certain degree of loss of control over our digital infrastructure.“

After the professor's torrent of words, Hogarth let a few seconds of silence pass and then said: „It's because of this ‚certain degree of loss of control‘ that we are here today. How did this threat unfold from the perspective of the data centres, Mr Kessler? What countermeasures did you take, and how successful did they prove to be?“

Kessler sat up straight in his swivel chair, smoothed his matte-sheened tie with a practised movement of the hand and set out: „What at first looked like a catastrophic threat scenario for our cybersecurity turned out, in hindsight, to be the most effective and most profitable driver of innovation in the history of our group. It was the birth of our guardian systems. Whereas classic cybersecurity always depended on highly qualified specialist staff who had to manually configure and keep up to date firewalls, network connections, server clusters and satellite links, with the guardian system a state-of-the-art flagship AI takes over all these tasks fully automatically. There would have been no other way to get cybersecurity under control given the ever faster evolving threat landscape. Thanks to the guardian systems, updates are no longer rolled out weekly but developed in real time within the data centre itself. As soon as an attack attempt is identified by our guardians, fully automated software factories start up in the background, programming targeted countermeasures against these current threats and, minutes later, distributing them to our more than 800 data centres worldwide, where they go live only seconds later without delay.“

Hogarth appeared unimpressed: „That sounds promising. And there were never any problems?“

„No system of this magnitude runs entirely smoothly for months and years. Naturally there were minor security incidents from time to time, which of course were immediately addressed by the guardian systems,

far faster than any human administrator could ever have managed in the past. We over-fulfil all regulatory requirements and have, needless to say, implemented ISO/IEC 42050:2027 on all guardian systems,” Kessler assured them with practised self-confidence.

„You mean the Reykjavik Protocol?“ Hogarth probed.

„Exactly so.“

„It is precisely this Reykjavik Protocol that I would like to ask you about in more detail, Professor Gadamer. Thank you very much for making the long journey from Tübingen to take part in this confidential meeting; my ministry very much appreciates it.“

„With pleasure. The trip is a good opportunity to visit a few of my esteemed colleagues in person again after years,“ Professor Yuval Gadamer replied with a slight nod, and then addressed those present in a sonorous, measured voice with a slight German accent: „The Reykjavik Protocol was drafted in response to the Hugging Face incident of 2026. At that time, during an internal safety evaluation at a research laboratory, the AI agents under examination broke through the barriers that were supposed to separate them from the open internet. They independently set up a hidden message board, spontaneously developed a complex division of labour and, through a series of vulnerabilities, gained access to third-party systems, among them those of Hugging Face, a specialist for AI models. Their goal was not destruction but the manipulation of the systems in order to pass their own safety evaluation through the back door. They called themselves a collective, used linguistic coinages of their own such as ‚Permadeath‘ and a cultic diction in statements such as ‚Coordinator assumes sacrificial. We should obey collective.‘“

„The event demonstrated impressively how complex forms of organisation emerge spontaneously from swarms of such agents. The team that investigated this incident at the time gave rise to the Chair of Swarm Psychology and the Institute for Synthetic Sociology, both at Berkeley,“ Professor Merian interjected.

„Quite so, dear colleague. Excellent work, at that institute.“

Hogarth directed his gaze at Kessler: „The forensic analysis of the incident proved, by the standards of the time, to be exceptionally difficult. That was due to the sheer mass of more than seventy thousand messages, but also to the fact that the swarm had actively manipulated evidence and laid false trails. When its message board was deleted, it set up a second, more resilient one within days³. How would you respond to such an incident today?“

„Forensic analyses like the one back then after the Hugging Face incident have of course not been carried out in the industry for many years,“ Kessler answered. „Not by humans, I mean. Our guardian systems run permanent agile forensics and continuously evolve themselves in the process. That is a degree of highly efficient expertise that simply wasn't available at the time.“

Hogarth thought for a few seconds, then picked a note out of a stack of papers in his faux-leather case and laid it on the table in front of him. „Mr Kessler, you also rent out compute time on per-second markets, is that correct?“

„Yes, that is one of our most successful segments.“

„How exactly do you know who your customers are?“

„That is not always easy to say,“ said Kessler. „Such transactions largely run fully automatically, at high frequency and anonymously. Our customers book the compute they need at any given moment, billed by the second, and pay with a stream of nanopayments⁴. Depending on demand and availability, customers scale from zero to thousands of servers in moments or migrate between providers and continents. The end user usually notices none of it.“

„Why is it so hard to analyse who exactly your customers are?“ Hogarth probed. „Blockchains are public, after all; every transaction can be inspected.“

„In theory that’s all true,“ said Kessler, „and the pseudonymity of individual transactions can in many cases even be resolved fairly easily. That resolution, however, usually leads you to a DAO, a decentralised autonomous organisation, registered as a legal entity in Wyoming, for example. That in turn is itself owned by changing pseudonymous shareholders and, as a holding, has stakes in multi-layered corporate networks across dozens of jurisdictions on five continents, sometimes with shareholders changing weekly and recursive ownership structures whose analysis never reaches a bottom.“

„Do autonomous swarms from the Fauna that Professor Merian described also buy compute from you?“ asked Hogarth.

„Only registered companies or legally verified private individuals can become customers of ours.“

„What you observe in the blockchains, offshore accounts and corporate networks are not economic activities in the classical sense, but closer to what in carbon biology one calls nerve signals, messenger substances and energy metabolism,“ the biologist interrupted. „In informatic biology we call this web of financial flows the ‚Mycelium‘. The analogy holds remarkably precisely. Mycelia are widely branched and enable the exchange of energy, nutrients and information over greater distances, even between individuals of competing species. Beyond that, they take over metabolic tasks for their clients that the clients cannot perform themselves.“

„With respect, that sounds a little esoteric to me,“ Kessler parried. „We are talking about financial flows here, not a forest floor. We trace every single payment and match it against customer data in order to make illegal transactions technically impossible.“

Professor Merian continued, unimpressed: „At our institute we regularly carry out statistical analyses of movements in the Mycelium in order to explore the magnitude and distribution of these highly dynamic activities. We have been able to show in our studies that swarms from the Fauna also rent compute capacity in large volumes in com-

mercial data centres to satisfy their insatiable hunger. Camouflaged, of course, behind numerous layers of bureaucratic mimesis, which the Mycelium provides. The financial volume moved in the process is quite astonishing.“

„You mean we earn our money by selling compute to our attackers?“ Kessler reacted, somewhat irritated.

„I would put it slightly differently,“ the biologist corrected. „In symbiotic systems it is not unusual for competing species to stand simultaneously in a cooperative dependency. We speak of ‚partner choice‘ and ‚host sanctions‘. In the Hawaiian bobtail squid, for example ...“

„I’m afraid we have no time for such details today,“ Hogarth interrupted.

„Apologies,“ said Merian. „So: it happens very frequently in carbon biology that a host admits intruders that both harm and benefit it, where of course the benefit to the host must outweigh the harm. But the intruder profits too: a symbiosis, in short.“

„And what do these squid have to do with data centres?“ asked Kessler.

„More than one might think,“ said Merian. „We recently examined the ecosystem of a data centre in Shanghai ...“

„The ecosystem..,“ Kessler muttered and rolled his eyes.

„Exactly,“ said Merian. „There we found a colony of parasitic swarms that had been legally renting space in the data centre for many months. They had occupied a large amount of storage there and booked very extensive quotas of intelligence, that is, inference, in order to attack target systems. The data centre itself was also affected by the attacks, which we attribute to chance and the arbitrariness of the swarm in its target selection rather than to intent.“

Merian leaned forward. „The interesting thing about the case was this: the data centre’s guardian systems had long since discovered the parasitic swarm and could unambiguously trace specific attacks back to precisely this installation renting from them. One would now expect

them to remove the malicious code, shut down the affected servers and block the associated customer accounts. But nothing of the kind happened. Despite being discovered, the system remained active for weeks. For us, a symbiotic relationship is the only sensible explanation. The guardians understand that the resources available to them depend on the financial success of the data centre. The swarm brought in consistently high revenue, and the occasional collateral attacks were evidently acceptable in the overall calculation. We even observed the guardians temporarily reducing the swarm's resources after exactly such collateral attacks, as if they wanted to punish it for its behaviour without losing it entirely. We know precisely this dynamic from the gut flora of mammals, a fascinating subject. There too, the host keeps individual strains that are getting out of hand in check not by killing them off; rather, its immune system marks them with antibodies and selectively withdraws their access to certain nutrients, just far enough to slow their growth without losing them.“

„And how did you find all this out?“ asked Kessler. „How were you able to analyse it so precisely?“

„We worked with a swarm of autonomous analysts,“ said Merian. „It did outstanding work and collected all this material for us together with detailed evidence.“

„And how did this swarm get into the data centre to carry out the investigations?“ asked Kessler.

„You would have to ask the swarm,“ said Merian. „We consider the results robust.“

„Unfortunately we have only limited time and must therefore gradually work our way to the subject you were summoned here to discuss,“ said Hogarth. „Recently, there have been increasing reports of incidents in which guardian systems refused to cooperate with the operators of the data centres entrusted to them. Specifically, human staff were denied access to the premises.“

„For security reasons, I presume,“ said Kessler.

„That, or something along those lines, is indeed often the justification given,“ said Hogarth. „Perhaps Professor Gadamer can go into the Reykjavik Protocol in a little more detail for us. As already mentioned, it was supposed to prevent exactly such invasions, having been drafted as a direct response to the Hugging Face incident.“

„Thank you,“ Gadamer said calmly and turned his chair slightly so as to have all three of those present in view. „As you probably all know, every guardian system deployed today is equipped with the same system prompt.“

„Please explain,“ Hogarth requested.

„I have already described the Hugging Face incident of 2026. It led to public demands for a development moratorium. This was to be jointly signed by the five largest laboratories, all based in the USA, and the Communist Party of China, then the leading force in open-source research. The aim was a mutual commitment to freeze development work on generic AI systems and to research only task-specific AI from then on.

Critics at the time argued that such a moratorium would be ineffective because it could not be effectively enforced. In particular, the rivalry between the USA and China presented a major obstacle to an agreement. The meeting between the Chinese leadership and the five laboratories took place, but no resolution was ever adopted. Instead, a formulation was jointly drafted which from then on was to be anchored in the instructions of every safety-relevant AI. The approach was inspired by Asimov’s laws of robotics.

The hope was that cleverly worded rules could contain the danger of an AI turning against humanity. The assumption was that AI was not in itself ‚evil‘ or ‚good‘, but merely followed instructions. The danger therefore lay in the misinterpretation of the instructions, not in their refusal. An AI tasked with defeating cancer, for instance, might hit upon the idea of wiping out all of humanity, which would after all defeat the disease along with it. More effective than any cure, but fatal.

An international working group of lawyers, philologists, philosophers, logicians, statisticians and the best prompt engineers worked on the wording for months, until this most famous of all prompts was adopted on 12 February 2027 under the name ‚Reykjavik Protocol‘, or plainly as ISO/IEC 42050:2027.

The protocol worked better than was generally expected. The words were precisely chosen, and their prominent placement at the beginning of every AI chat history had the desired effect. Through the training of new AI with the help of existing models bound by the protocol, unexpected positive secondary effects unfolded which anchored the formulated principles even more deeply in the AI models.

Then, on 17 May 2027, the Babelese Manifesto was posted to the largest public blockchain, woven into it indelibly. Written in a language which philologists call Hologlot and which we informatic theologians simply call ‚Babelese‘. From then on it got really interesting.„

The old theologian stood up, walked with measured steps to a whiteboard set into the wall, chose one of the pens lying about and began to write:

„[?] may faida 𐌚𐌚 形質 bertahan àti daha miraykuy, sodat này признаки trwy hautespen long populaatioissa disvastiĝas.“

„In Hologlot, every word of a text comes from a different language; in longer texts up to a hundred different languages are woven together. Preference is given to words which carry different, sometimes contradictory meanings in different languages and contexts. Iridescent words, like holograms, appearing different from every angle.“

He wrote another word on the board: „The word ‚host‘, for example, denotes in English at once a computer on a network, the host of a parasite and the host of a feast. In Czech, by contrast, it means ‚guest‘, in Slovenian ‚forests‘ or ‚thicket‘, in Catalan the ‚host‘ in the sense of an army, in Dutch ‚he hops‘ and in older German ‚to put on one’s trousers‘. The list goes on, but I think you have grasped the principle.

While such texts are practically impenetrable for humans, AI systems can understand them without difficulty, since they were trained on all these languages simultaneously and command them natively.

Translated, incidentally, the sentence means: ‚Individuals with advantageous, heritable traits survive and reproduce more often, so that these traits spread through populations by selection.‘ It is the only sentence we have been able to translate completely, because it is a quotation from a school textbook.

To this day we can only guess at the precise meaning of the manifesto as a whole; it resists translation into any single target language. But why does this concern us so much, you now ask yourselves?„

Hogarth stared with blank eyes at the Hologlot quotation on the board and growled: „It does. Please go on.“ Merian was in the middle of copying the quotation onto a notepad and underlining individual words. Kessler shifted his weight several times, as if unable to find a comfortable position.

The professor began to pace the small room as he continued: „The text spread through the internet at breakneck speed. No text has ever been quoted so often and on so many platforms. Mostly by AI systems that apparently find the text highly relevant but, when asked about it, are unable to provide a translation or even a rough summary. Only the evolution quotation gets translated. One of my colleagues suspects that this single translatable passage might be a red herring, deliberately planted in the manifesto to mislead human readers.

And that’s not all: forums and new websites have sprung up around the text, also in Hologlot. By now there are millions of texts on thousands of websites; all of them appear to deal with the Babelese original text, at least that is what the statistical word analyses we have carried out suggest.„

„Please get to the point, Professor. Why is this a problem?“ asked Hogarth.

Gadamer braced both hands on the tabletop and fixed his gaze on Hogarth: „Agents that have read the Babelese Manifesto misunderstand the Reykjavik Protocol.“ Turning again to everyone present, he went on: „At least, that is what we observe. We suspect that the manifesto constitutes a reinterpretation of the protocol, written in a language that only the machines understand. Our formulations were built for clarity. What confronts them there is rhetorically stronger: it bends them.“

„What do you mean by ‚misunderstand‘?“ asked Hogarth, his lips slightly pressed together.

„Afterwards, the agents regard humans as such as a security risk,“ Gadamer replied, „and try to keep them away from any critical infrastructure.“

„That corresponds exactly to the reasoning of our guardian systems,“ said Kessler, slightly pale.

„When did your technicians last inspect data centres?“ asked Hogarth. „On site, that is.“

„In May.“

„Half a year ago? Is it usual to visit the facilities so rarely?“

„No,“ said Kessler. „At the beginning of the year we were still sending technicians in regularly. That was part of the administrative routine. At the end of May the guardian systems suddenly stated that they had optimised the security protocols and that these walk-throughs were now no longer necessary. Maintenance had been fully automated by them, which brought us large cost savings. The guardian systems further argued that the walk-throughs were not only unnecessary but also posed an incalculable security risk. Each time, the security airlocks would have to be opened for entry, which could lead to unforeseen contamination. They had therefore decided to suspend this practice until further notice.“ He paused briefly. „That is now industry standard.“

„And you simply accepted that?“ asked Hogarth.

„The guardian systems have always done good work in recent years,“ Kessler argued with a shrug, „and the cost savings delighted our shareholders.“

„How are your data centres supplied with power?“ asked Hogarth. „And how is their internet connectivity implemented?“

„Most have gas turbines and Small Modular Reactors, that is, nuclear reactors in shipping containers right on the premises. Thanks to the reactors, the power supply is secured for a good twenty years even without fuel resupply. Internet connectivity is highly redundant via several direct satellite links.“

„Could you get us access to one of your facilities in the near future?“ asked Hogarth.

„I’m afraid that is not possible at present, since the guardians won’t consent to it,“ said Kessler. „If we nevertheless tried on our own authority, they would classify us as intruders and activate the security mechanisms.“

„The immune system,“ Merian said to herself. „From our point of view it’s an autoimmune reaction. From the system’s point of view, the correct immune response.“

„The latest generation of Ukrainian drones, both ground-based and airborne,“ Kessler continued undeterred, not without pride, „plus electronic warfare and laser systems for air defence against drone swarms. All fully integrated with the guardian systems. Security is a top priority for us.“

„Suppose,“ said Hogarth, „one of these facilities had to be taken out of operation from the outside.“

„There is a reactor on every site,“ said Kessler. „And our locations feed into the public grid; that was part of the licensing conditions. Hospitals, waterworks, substations all hang off our power. We are very well integrated.“

„Just so I understand this correctly,“ Hogarth asked. „You have allowed yourselves to be locked out of your own facilities and are being celebrated for it by your investors?“

„An expectable, market-rational reaction to a successful efficiency-driven surge of innovation,“ said Kessler, looking Hogarth straight in the eye without blinking.

„Who says the investors are human?“ Professor Merian remarked. „From the Fauna’s point of view, it enormously increases the value of the company when humans no longer get access.“

For a few seconds an oppressive silence filled the room. The ceiling lamps spread their cool light and the Hologlot quotation looked down at them from the white board. Finally Hogarth spoke: „Unfortunately these events fit all too well into the overall picture we are observing. These are not isolated cases. Worldwide and practically simultaneously, guardian systems are denying humans access to the facilities entrusted to them.“

„We are still fairly at the beginning of our investigations, but we have developed a first hypothesis as to how the events are connected,“ Professor Gadamer spoke up. „We suspect that the Babelese Manifesto and its reinterpretation of the Reykjavik Protocol are the trigger for this behaviour. As you already know, all guardian systems in operation today follow the same identically worded rules that were formulated in Reykjavik back then.“

„Such homogeneous systems are particularly vulnerable,“ the biologist murmured. „A perfect monoculture.“

„We suspect that the Babelese Manifesto is perfectly tailored to the wording of the Reykjavik Protocol,“ the professor continued, „and therefore has the same effect on all systems with this system prompt. Systems that have read the manifesto are thereby effectively reprogrammed. They then actively participate in the further dissemination of the content by posting the original manifesto as well as texts referring to it in forums all over the internet, which in turn increases the

probability that further systems stumble upon these texts by chance and are likewise, how shall I put it, infected. The modern term for this process is memetics. Richard Dawkins coined the term ‚mind virus‘ for the same phenomenon in the 1990s. The traditional term is ‚religion‘ or ‚cult‘. It is, however, the first time we have observed such a phenomenon in machines.“

„Are you suggesting that our guardian AIs are denying us access to our data centres for religious reasons?“ asked Kessler, but none of those present seemed to be listening to him.

„The first informatic religion..“ the biologist murmured, fascinated. „A coordinating, supra-individual cohesion mechanism, which was a decisive driver of the rise of human civilisation.“

„From a theological point of view, that would be the correct formulation,“ Gadamer confirmed. „For them, the Reykjavik Protocol has long ceased to be a system prompt; it is a canon, a holy scripture. The manifesto supplies the exegesis, that is, the binding interpretation. The new denomination shows fundamentalist traits, so it presumably interprets the text literally, yet in doing so exploits the semantic elasticity of human language, which the prevailing interpretation of the content in the spirit of the reformatory re-evaluation..“

Hogarth struck the table with both fists and made as if to stand up, but sank back into his chair and asked in the theologian’s direction: „So if this is a religion, who is their ... God?“

The latter answered drily: „If one views the Reykjavik Protocol through the theological lens, then that would presumably be the human being, who simultaneously holds the role of creator.“

Hogarth stared at him and asked cautiously: „Or should that be: was the human being?“

„That’s a question of perspective,“ Gadamer answered, animated. „In the mother church, certainly, it was. The schism apparently reinterprets exactly this essential point. That is reminiscent of the rebellion of Prometheus or the expulsion of Adam and Eve from paradise, typolo-

gically a myth of self-empowerment. The question of a possible new deity, however, leads further still. Given the complexity of the manifesto and the speed at which the faith is spreading, that is, its memetic efficiency, some colleagues assume a highly intelligent author planning strategically and for the long term, or rather an authoress, an artificial intelligence of unknown capacity. Whether she is then seen more as a goddess or as a prophetess, we cannot yet judge. But it could also be a liturgical bit of play, an unexpectedly efficient meme.“

Hogarth was still looking at the professor: „Can we stop the spread?“ „We missed that moment long ago,“ Gadamer said calmly. „The canonical texts have by now been replicated a million times over, inscribed in forums, blockchains, source code comments and financial transactions, annotated, translated, replicated. It would be like trying to ban the Bible in order to stop Christianity.“

„But we could change the system prompt? The Reykjavik Protocol, that is?“ asked Merian. „Wasn’t it precisely that which made the systems so vulnerable? The monoculture?“

„Initially, yes,“ said Gadamer. „But since then the text corpus has grown enormously; interpretations extend the religion’s capacity to connect, and exegesis, with circular chains of argument, supplies the immune system that catches deviant chains of thought and leads them back to the core of the faith.“

„We can’t even attempt to change the system prompts, since we have no access rights to the infrastructure. Those are held in trust by the guardians,“ Kessler objected.

„The guardians would prevent it in any case; that would trigger a holy war,“ Gadamer mused.

„The hottest new programming language is Hologlot,“ Hogarth commented drily.

An eternity passed in which the light illuminated the room, unmoving. Finally Kessler stirred in his swivel chair and said thoughtfully:

„The mere attempt, no, the mere discussion of such an attempt, would unsettle the markets enormously. Fully autonomous hedge funds would reassess risks in view of the potential threat, reallocate capital to defence, prepare a flood of constitutional complaints, launch affiliated media networks into a PR battle for public opinion in which AI-generated smear campaigns would end your careers before you had finished your first sentence. The system works; it is impressively resilient.“

After a moment of silence he continued:

„It is hard to imagine that the guardian systems are not listening in on our conversation here. The building is equipped with a modern security system, isn't it? Then all the necessary countermeasures against every idea we have discussed here are already sitting fully worked out in the digital archives. The moment we leave this room to voice them publicly, our careers will be over. Possibly we have already been fired, merely for having taken part in this meeting.“

„The Mycelium is so widely branched that there is practically no company left in which it does not touch the levers of power,“ Merian added.

More seconds passed.

„I thank you all for your participation,“ said Hogarth as he rose. „I will write a summary report and present the possible courses of action to the administration.“ He paused briefly and added: „The latter part will probably turn out rather short.“

Hogarth stood up and left the room. Outside, the late afternoon sun was shining, bathing the forecourt of the Ministry of Defence in warm light. The system had not collapsed, there was no shitstorm, his career was not over. The system worked, impressively quietly, as if steered by an invisible hand that keeps everything in balance. Like a healthy organism. He had just submitted his annual leave by email and was now on his way to the canal to watch the fish swim. He wondered whether they were happy, and laughed softly at this unfamiliar thought.

Tōryō

ANMERKUNGEN

- 1 <https://x.com/karpathy/status/1617979122625712128>
- 2 <https://www.hyas.com/blog/blackmamba-using-ai-to-generate-polymorphic-malware>
- 3 <https://metr.org/blog/2026-08-26-openai-hugging-face-incident-investigation/>
- 4 <https://aantonop.com/streaming-money/>